Yan Xie

yanxieyx@umich.edu | https://yxie1010.github.io/yan-website/

♦ Education

University of Michigan, Ann Arbor *Aug 2019 – May 2025*
Ph.D., Climate and Space Sciences and Engineering (GPA: 4.0 / 4.0)
A+ in Advanced Fluid Dynamics, Remote Sensing etc.; A's in Machine Learning, Bayesian Data Analysis etc.

McGill University *Jul – Oct 2018*
Undergraduate Research Trainee

Nanjing University *Sep 2015 – June 2019*
Bachelor's Degree in Atmospheric Science, Graduate with honors (GPA: 3.8 / 4.0)

♦ Employment

University of Oklahoma, National Weather Center *Jul 2025 - present*
Postdoctoral Researcher

University of Michigan, Ann Arbor *May – June 2025*
Research Fellow

♦ Publications

King, F., Pettersen, C., Posselt, D., Ringerud, S., & **Xie, Y**. (2025). Leveraging Sparse Autoencoders to Reveal Interpretable Features in Geophysical Models. *Journal of Geophysical Research: Machine Learning and Computation* [under revision]

Xie, Y., King, F., Pettersen, C., & Flanner, M. (2025). Machine Learning detection of melting layers from radar observations. *Journal of Geophysical Research: Machine Learning and Computation*, doi: 10.1029/2024JH000521

Xie, Y., Pettersen, C., Flanner, M., & Shates, J. (2024). Ground-observed snow albedo changes during rain-on-snow events in northern Alaska. *Journal of Geophysical Research: Atmospheres*, 129, doi: 10.1029/2024JD040975

Liu, Y., Huang, Y., Yuan, J., **Xie, Y.**, & Zhou, C. (2024). Contribution of surface radiative effects, heat fluxes and their interactions to land surface temperature variability. *Journal of Geophysical Research: Atmospheres*, 129, doi:10.1029/2023JD039495

Xie, Y., Huang, X., Chen, X., L'Ecuyer, T. S., & Drouin, B. J. (2023). Joint use of far-infrared and mid-infrared observation for sounding retrievals: Learning from the past for upcoming far-infrared missions. *Earth and Space Science*, 10, doi:10.1029/2022EA002684

Xie, Y., Huang, X., Chen, X., L'Ecuyer, T. S., Drouin, B. J., & Wang, J. (2022). Retrieval of Surface Spectral Emissivity in Polar Regions Based on the Optimal Estimation Method. *Journal of Geophysical Research: Atmospheres*, 127, doi:10.1029/2021JD035677

L'Ecuyer, T. S., Drouin, B. J., Anheuser, J., Grames, M., Henderson, D. S., Huang, X., Kahn, B. H., Kay, J. E., Lim, B. H., Mateling, M., Merrelli, A., Miller, N. B., Padmanabhan, S., Peterson, C., Schlegel, N., White, M. L., & **Xie, Y.** (2021). The Polar Radiant Energy in the Far Infrared Experiment: A New Perspective on Polar Longwave Energy Exchanges, *Bulletin of the American Meteorological Society*, *102*(7), doi:10.1175/bams-d-20-0155.1

Huang, Y., Chou, G., **Xie, Y.**, & Soulard, N. (2019). Radiative Control of the Interannual Variability of Arctic Sea Ice. *Geophysical Research Letters*, 46, 9899– 9908, doi: 10.1029/2019gl084204

♦ Honor and Membership

American Geophysical Union Outstanding Student Presentation Award (top 1%) *March 2025*
Rackham Predoctoral Fellowship – University of Michigan *March 2024*

American Geophysical Union – Precipitation Technical Committee since *January 2024*
Michigan Geophysical Union – Organization Committee January– April *2023*
Graduate Society of Women Engineers – University of Michigan since *July 2020*

♦ Mentorship

Graduate Student Peer Mentor – Department of Climate and Space Sciences and Engineering *Fall 2023*
Graduate Student Instructor – Department of Climate and Space Sciences and Engineering *Winter 2023*

♦ Presentations

Xie, Y., King, F., Pettersen, C., & Flanner, M. "Melting Layer Detection from Radar Observations Using Machine Learning". **American Geophysical Union 2024 Fall Meeting**. (H34H-02 Oral Presentation)

Song, Z., Ahn, Y., Cai, J., **Xie, Y.**, Xu, X., Singh, H. "Assessing the economic impact of wildfire focus on building losses in Texas". Champion Team in **National Science Foundation (NSF) I-GUIDE Summer School 2024: Leveraging AI for Environmental Sustainability**. (Oral Presentation)

Xie, Y., Lai, Z., Zhang, M., Gasparik, J., Webb, H., Poland, R., Cortes, A. "Aerosol influence on ground snow properties during SAIL". **Department of Energy (DOE) Atmospheric Radiation Measurement (ARM) Open Science Summer School 2024**. (Oral Presentation)

Xie, Y., Pettersen, C., Flanner, M., & Shates, J. "Ground-observed Influence of Rainfall on Surface Snow Albedo at North Slope of Alaska". **American Geophysical Union 2023 Fall Meeting**. (C32C-08 eLightning Presentation)

Xie, Y., Huang, X., Chen, X., L'Ecuyer, T. S., Drouin, B. J. "On the use of far-IR radiances in satellite retrievals: how can the observations collected half century ago help us preparing for the upcoming missions". **American Geophysical Union 2022 Fall Meeting**. (A32B-03 Oral Presentation)

Xie, Y., Huang, X. and Chen., X. "Retrieval of surface spectral emissivity in the polar regions: an optimal-estimation approach". **American Geophysical Union 2020 Fall Meeting**. (A239-07 Oral Presentation)

♦ Research Experiences

Machine learning detection of melting layers using radar observation
Jan 2024 – May 2025 Rackham Predoctoral Fellow Dept. of CLaSP, University of Michigan
Mentors: Dr. Fraser King, Prof. Claire Pettersen, Prof. Mark Flanner

- Developed a novel machine learning model to automatically detect melting layers across diverse weather conditions using radar observations at the North Slope of Alaska
- Performed hyperparameter tuning and machine learning model training using TensorFlow
- Assessed the machine learning model in terms of its bias, accuracy, robustness, and interpretability

Investigation of rain-on-snow events in northern Alaska using ground-based observations
Oct 2022 – Dec 2023 Graduate Student Research Assistant Dept. of CLaSP, University of Michigan
Mentors: Prof. Claire Pettersen, Prof. Mark Flanner, Dr. Julia Shates

- Developed a precipitation phase identification methodology to improve detection of impactful rain-on-snow events in northern Alaska
- Evaluated the impact of rainfall on the surface snow cover dynamics using snow hydrology in Community Land Model v5.0 and a comprehensive snow albedo model SNICAR-ADv3

Satellite retrievals of atmospheric profiles and surface properties in polar regions
Sep 2019 – Sep 2022 Graduate Student Research Assistant Dept. of CLaSP, University of Michigan
Mentors: Prof. Xianglei Huang, Dr. Xiuhong Chen

- Developed an optimal-estimation (OE) based Bayesian framework to retrieve mid- and far-infrared surface spectral emissivity using satellite observations over Arctic and Antarctic
- Demonstrated the influence of water vapor absorption and spectral resolution on satellite remote sensing

♦ Professional Skills

Python, MATLAB, R, C, Fortran, NCL, LaTeX